

"Physics as Information Processing" ~ Chris Fields ~

Lecture 1

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hello and welcome everyone to the active inference Institute this is session one of the course physics as information processing with Chris fields first we'll have Andre Aguirre and Chris Fields introduce themselves and then we'll carry on with the first lecture here check out the video description for a link to the course overview website where you can ask questions that will be answered asynchronously register to participate in the discussions which happen about two weeks after each of the six lecture sessions and just learn more about this area so thank you both so much for joining into this adventure we are starting now and first please Andre Aguirre introduce yourself and then Chris introduction and lecture thank you uh hello so I'll be the course assistant and um specializing in probability and I have a deep interest in the physics of information and I've been uh familiar with Chris's papers for a while yeah so yeah just here to learn myself uh thank you Andrew and thank you Daniel um I'm Chris fields and uh I'll be presenting this course in six sessions and uh Andrew will be organizing discussion sessions after each of those and all of this is explained on the course website so uh let's start this is a course on physics's information processing and this first session will be a historical perspective on the idea that physics is is or is about information processing and I'll just start with a few quotations that span the middle of the 20th century from Wittgenstein and 1920 saying the world is all that is the case so defining the world in terms of facts not objects a land hour in the early 60s proclaiming that information is physical and then uh John Archibald wheeler who in many ways is the grandfather of this era recently um stating it is it from bit so uh things come from information I.E bit strings and if nothing else this shows that that formulations of this idea get pithier as the 20th century rolls on but the history goes uh back further clearly than the 20th century that I'm only going to really talk about a piece of it um the timeline that I'll actually discuss today the the most relevant history of this idea goes back to the mid 19th century and the first specific thing I'll talk about is clausius's definition of entropy but with the beginning of the understanding of thermodynamics and the role of

information in thermodynamics you get this very interesting multi-disciplinary progression of ideas that incorporates the beginning of quantum theory and the the beginning of quantum theory can kind of be dated to the the first Solvay conference in 1928 and the famous debate between Boren Einstein over whether quantum theory is about knowledge information or objects things uh but it incorporates a lot of work in computer science and logic and Mathematics so interestingly computer science was born effectively in the mid 30s with the work of church and Turing which very rapidly converged with the work in physics so today we'll be talking about both computer science and physics and then in the second half of the 20th century this just exploded into a huge area and in consequence of that mid 20th century development we're beginning to see a new idea about physics which is roughly encapsulated in Quantum information Theory and the new idea is this it's that what physics is actually about is information transfer across boundaries and the information we can represent the information transfer like this and this is a conventional use a boundary is always a blue ellipse and the Agents that are exchanging information across this boundary are conventionally called Alice and Bob which is just a more polite way of saying A and B and when you think about uh this picture it becomes clear that what physics is really about is communication and this is a a wild redescription of the idea of what physics is compared to of the ideas of Newton or LaPlace or even the 19th century ideas and it's very different from the idea that's been preserved in 20th century physics in the lineage of Einstein and others who viewed classical physics as in a sense of fundamental um either as completely fundamental or as a fundamental adjunct to quantum theory so this way of thinking about physics is a very deeply Quantum theoretic way of thinking about physics and where we're going in this course uh today is really uh the how did this all happen it's the origin story and then in the next session I want to discuss Quantum information Theory explicitly and in particular how quantum theory makes this conclusion that physics is about communication very simple and obvious much more obvious than it is in classical physics where it takes work to to formulate this idea uh then in the next session we're going to talk about semantics and how observations become meaningful to the agents who make them and hence how actions become meaningful then in August we'll talk about uh communication Theory a little bit more and talk about how agents employ multiple communication channels when they're communicating and you know this is this is obvious when you think of people communicating they not only talk to each other they look at the same things they point to things Etc so this is what I mean by then in September we'll leverage that discussion to talk about how space-time actually emerges from communication and this is

one of the most important aspects I think upon an information Theory it provides us with a way of viewing space-time as an emergent phenomenon that communication is what is fundamental in some ontological sense and the Box in which it happens face time is not so the final session in October will talk about applications to biology via the free energy principle and future directions both in physics and biology and elsewhere so it's going to be an interesting ride I'm keeping formalism to a minimum uh because we're directing this toward the broad array of people who are interested in active inference and who are involved with the active inference Institute and I would ask you to hold questions because we have a lot to get through in an hour for the interactive discussion and for the discussion forum um so uh I hope I explain things well enough that all of the concepts will be understandable uh if not Wikipedia is actually a wonderful resource in this area for just definitions of terms so if there's anything that that just a term that is is a trip up uh try Wikipedia it's probably a very good source for what these terms mean so let's start um our story as I said begins in the 19th century and in the mid 19th century lots of physicists were devoting their efforts to figuring out how to make better steam engines uh one question that arises when you're trying to design a steam engine is what happens physically when you add heat to a system at constant temperature so if you're building a steam engine you've got a boiler because you need to make Steam and as you turn up the heat to your boiler you get more steam but the temperature doesn't change so this is a mystery uh what what is the heat actually adding to the boiler that is not increasing the temperature clausius responded to this question in a way that's sort of typical for a physicist or a mathematician since he didn't know what the answer was he just invented a new name for whatever it was and gave it a formal definition so he called it entropy which is a made-up word that if it was translated from the Greek would roughly mean transformation content in Trophy and he represented it by a simple equation that the change in this uh New Concept entropy which is always called s is just equal to the change in heat Q at constant temperature t so this obviously this equation just reformulates the question in declarative form saying whatever this stuff is uh it's it's changes in this stuff are just changes in heated constant temperature well heat is energy and this wasn't completely recognized in the mid 19th century but the way you'll see this equation in a current textbook is DS is the it energy change in energy at constant temperature so even more commonly you would see it written as the change in energy is equal to the temperature times the change in the entropy it's the most common sort of textbook way of seeing this but the question of course is what is this quantity what is this entropy what does this concept mean and about 15 years after clausius proposed it boltzmann

had the key insight which is that entropy is a measurement of our uncertainty about the state the system is in and in particular oh he again of course went to formalism and said the entropy S is equal to some constant k times the number of states that the system can be in that look the same to us and since that number of states is enormous the way to make that manageable is take to take the log of the number of states so the natural log is $\ln$ and this constant k is called Boltzmann's constant boltzmann was able to do this because he subscribed to a radical and very unpopular Theory of that material things uh including gases like air were made of atoms and heat made the atoms move around and as you increase the amount of heat the atoms can move in many different ways so the number of states that they can be in that look the same to us increases and that's what entropy is it's this increase in the number of states that the system can be in uh that all look the same to us with the measurements that we can make and since they look the same to us we're uncertain about exactly what state they're in so so entropy is a measure of uncertainty this was really the beginning of modern oh because what it says now is that decreasing uncertainty requires energy it links a measurement of uncertainty to a measurement of energy and if you think about the uncertainty principle in quantum theory how the core idea of the uncertainty principle is you can't measure a system without disturbing it so to actually act on a system requires energy and that's what you have to do to get information so here's boltzmann basically inventing so we're going to fast forward by another 15 years to 1900 and in 1900 Planck solved this problem called the black body radiation problem which was basically how much heat does your hot boiler give off into the air and of all of the measurements of the heat that hot boilers gave off to the air ran into problems in classical physics and caused contradictions and quantities that went to infinity and all that was bad so many people were trying to solve this problem and Planck solved it by making a simple postulate he said the energy of the radiation is proportional to its frequency so its color in the case of light and if you go higher frequency you end up in ultraviolet and x-rays and gamma rays if you've got a lower frequency you go into microwaves and radio and all of that so this is a nice way of talking about radiation and it turned out that this solved the problem I mean just assuming this simple proportionality relationship uh produced Spectra for black body radiation that worked that matched what you saw experimentally well this means something very important it means because this number h the proportionality constant called Planck constant is a number it's finite it means that energy comes in discrete units of h you can have one h or two h or 10 million h but you can't have half an h of energy so it's quantized and this is widely recognized as the birth of quantum but of course we should have known this already if

we just thought a little bit right we know that that changes of energy are proportional to the changes in entropy uh by temperature and we know that that entropy is a measure of the number of states and numbers of states are just numbers you can have one state two states three states 10 million States 100 billion States but they're all just a number uh one up and it's not infinite there's not an infinite number of states unless you have an infinite amount of energy which you don't have so we knew already that entropy could only take discrete values and since energy and entropy are basically the same thing we knew already that energy could only take discrete values so we could have realized in 1900 that energy is quantized because the number of states is quantized uh so it shouldn't have really been a mystery why energy was quantized but it was a mystery and it stayed a mystery it's I mean it's in a sits still a mystery people still debate the meaning of quantum theory but another thing we could have known in 1900 was something very important and it's that this Quantum of action of Planck's constant which has units of action which is energy times time is intimately related to Boltzmann's constant uh and Boltzmann's constant has units of energy over temperature but this wasn't actually understood until the 1950s no one really figured that this out there was this relation until the 1950s and when it was figured out it was figured out by a guy named Giancarlo Wick and he introduced this notion of the wick rotation by realizing that if you have an equation in classical physics and in it there's the term $1 \text{ over } KT$ you can always replace that one over KT with this other expression $i \text{ t over } \hbar$ $\hbar$ is just h divided by 2π and you'll get an equation that's valid in quantum theory and this is typically described in textbooks as a trick and whenever something in physics is described as a trick what that really means is it's something we don't understand uh and lots of papers have been written about the meaning of the wick rotation um but to start to understand the wick rotation I want to look at this equation a little bit of $1 \text{ over } KT$ is $1 \text{ over } \text{energy and } I \text{ T over } \hbar$ is 1 over an energy since $\hbar$ is units of energy times time uh we have time in this equation and then we have this factor I which is typically thought of as just an imaginary number so it's the square root of -1 and what's this I doing in this equation and in fact you see factors of $i \text{ t over } \hbar$ if you're familiar with quantum theory everywhere so what's the meaning of this imaginary number if you just think of $i \text{ t}$ or of i as this arbitrary imaginary number that somehow renders the equation mysterious then the whole whole of quantum theory is mysterious but the wick rotation is very mysterious but what I is actually is an operator and if you think of the real numbers as an axis which is always drawn horizontally so here it is here's zero and the axis is pointing that way I'm sorry I can't get the camera really far enough away to see my arm here

so what is i i is actually an operator that rotates the whole real axis by 90 degrees so if you if you see a plot of complex numbers and the real numbers go this way and the quote imaginary numbers go that way so what multiplying by i is done is rotate by 90 degrees and of course if you do i squared you rotate twice by 90 degrees my arm will do this but you end up pointing that way and those are the negative numbers so i as an operator that rotates something by 90 degrees and if you rotate four times by 90 degrees you're back to the identity so i to the fourth is one so this tells us something very interesting which is that what the wick rotation is really talking about is it's a geometrical equation and in July we'll come back to this and really probe what this Wick rotation means physically um but as I said this wasn't understood to the 1950s and by the 1950s a lot had happened so quantum theory have been developed boring Einstein had had their debate particle accelerators had built the atomic bomb had been built Nuclear Physics as well in its way um so an enormous amount of practical physics had been done quantum theory was highly developed people were starting to think about Quantum field Theory before they made this this simple realization that's formulated in the wick rotation so this is a heartbreaker of things to come but before we continue in physics we need to backtrack and time a little bit and look at what the mathematicians were doing so across the hall in the math department one year after the Solvay conference in 1929 uh Kurt girdle proved his famous first in completeness theorem and the theorem states that no formal system that contains arithmetic can be both consistent and complete and that means that either there are true statements that aren't provable in the in the formal system or there are false statements that are provable or of course both and girdles proof is is actually extremely simple almost all of the work in the proof is setting up all of the notation and procedures and so forth to formulate within arithmetic uh the sentence this sentence is not provable and once you have that sentence formulated within arithmetic then the conclusion of the proof is is obvious you know if you can prove the sentence this sentence is not provable then you've proved something that's false and if you can't prove it then there's then there's sentences that you can't prove uh in arithmetic so this was incredibly bad news for mathematicians who thought that finite discrete operations which is what proofs are and also what computations are can exhaustively enumerate the facts and this was the Assumption behind wittgenstein's claim that the world is all that is the case the world is a collection of facts and optimistically he thought that first order logic would allow us to enumerate all those facts and we'd be done dreams of a final Theory again um so so girdles there means that no system with finite capabilities no system that can just do find out discrete operations can fully describe its environment it will

always be in an environment where there are two things that aren't provable or false things that are provable but I think more relative to a discussion of Agents is that it means that no agent can describe itself uh any agent's theory of itself will either contain true statements that it can't derive or false statements or it will either mystery statements that it can't derive or it will end up deriving things that are false about itself of course we see this in Psychology all the time so an immediate consequence of Godel's theorem was an intense investigation of what computation actually is what it meant to talk about finite discrete operations and two leaders of this were of course Church and Turing and here's a picture of a Turing machine which is just a little device with a couple of tapes and a tape reader and a simple logic unit that either writes or one or a zero if it sees a one or a zero and they defined a computation as a process that can be implemented in finite Time by such a machine or by Church's Lambda calculus or by any of the now hundreds of other methods that are provably equivalent to a Turing machine so what does this mean it means that computation is a physical process that can be mechanized and it turns out mechanized in any one of a huge array of ways it means that many different implementations of any computation are possible so I can do it on a Turing machine I can do it on my laptop I can do it in my head Etc the most important thing it means is that there are questions with no computable answer this is the Revenge of Godel's theorem and two of the most famous questions of this kind are given some arbitrary program will it halt will it will it get to an answer in finite time and that the answer to that question is this is undecidable no procedure can figure this out and the other the other undecidable question is given some arbitrary program what function does it compute and you'd think that would be simple that you can read a program and figure out what function it computes but it turns out that is undecidable that cannot be done by any finite process so this was another body blow to the goal of understanding everything with uh finite discrete processes but it also set the stage for something new it set the stage for thinking about an agent who interacts with a computational process by giving it an input and then looking at its output sometime later and this of course will look familiar because I've included the blue ellipse which is the boundary which these days we call a user interface and the user interface just allows some finite action on the system and then the ability to observe some finite response by the system so we can now ask what can Alice determine by acting in some finite way and then making some finite number of observations I.E. receiving some finite number of outputs from the system that she's acting on and the first 20 years of this produced a large number of answers all of them negative so to go back to Turing he proved that Alice can't tell what's implementing the

function that she sees being implemented uh she can't tell whether a given input will lead to an output that's the halting problem Oh Shannon showed that Alice can't tell what the inputs mean to the system of his whole theory of communication is completely independent of semantics and his theory of communication actually accurately describes what Alice can observe uh rice is the one who proved that you can't determine what program is generating the outputs and then Moore proved a very similar result in a completely different formal setting of general cybernetics that you can't tell what process is generating the outputs by finite observation look what Alice can do is build a predictive model of what generates the output she sees in response to her inputs and test it by designing new inputs and this is of course as as Carl popper told us the process that we call science so owls can do science even though she can't answer any of these fundamental questions now this of course has a huge technological consequence uh since this theory of computation tells us that processes are effectively virtual uh we we don't know what they are and we can't determine what they are except in theory by making the theory uh technologically it means we're free to use virtualization everywhere because we have to deal with it anyway and this allows us to build multi-level architectures it means that we can architect computers where no layer of the computation has any idea what's going on below or above it and doesn't need to uh and that's what makes practical programming possible so from these no-go theorems that tell you what you can't do you actually get an enormous boost into it and use to probe the world and develop theories and on and on and on so in a sense girdle birthed not only computer science but practical Computing by showing us that virtualization is just the way the world works so now let's go back to physics where these ideas were were replicated basically reintroduced reinvented by Feynman in developing his path integral formulation of quantum theory and basically what Hyman realized was that in any physical process of The Observer Alice prepares some state that she's interested in she prepares some input to an experiment then she lets something happen and then she sees what the result of the experiment is and the canonical experiment in physics is scattering you you you know fire two protons at each other and they intersect someplace and stuff comes out and you measure this stuff that comes out what you can measure is momentum and spin energy things like that position these processes conserve the total values of the things you can measure so in particular they preserve momentum and angular momentum and they preserve you know other things that are harder to measure and are only approximately conserved anyway like lepton number uh but feynman's contribution to this way of thinking about experiments was to say look if you want to understand the output you have to sum over all of the possible processes

that could have produced the output from the input no matter how improbable they are so the famous idea for refinement diagrams is if you have an electron that's scattering off an atom oh you measure the initial state of the electron that you've generated with an accelerator or something and you measure the final state which involves momentum and spin and so forth and for all you knew uh part way through the process the electron disappeared and an entire universe appeared and then annihilated with a copy of an entire anti-universe and the electron came back out uh and you have to include processes like that if you really want to understand and correctly predict the outcomes of your measurements and uh Murray Galvan lifted does the bumper sticker slogan from *White and The Once and Future King* everything not explicitly forbidden is mandatory that sums up feynman's idea and this is called the totalitarian principle since it's what's written in *White's* book on the kingdom of the ants for which everything not explicitly forbidden is mandatory so let's think about a real example uh that's a bit uh actually it's the same as scattering the ultimate scattering experiment uh in physics is a black hole uh stuff goes into the Horizon of stars whole galaxies whatever something happens and stuff comes out and what comes out is hawking radiation uh information is conserved if the information in the Hawking radiation is actually the same as the information and the stuff that went in and conservation is not qualitative it's quantitative it's only the quantity of information that's supposed to be preserved and you can ask how much is the information that's being emitted by a black hole that's emitting Hawking radiation and beckenstein was the one who figured this out using an incredibly simple argument but I'll just give the answer here the answer is that the total entropy of the black hole is its area divided by four and the area in this equation has to be computed in Planck units which are units where Planck's Constant and the speed of light in Boltzmann's Constant and other interesting things are all set equal to one so one plank area is the plank length squared which turns out to be about 10^{-70} meters squared so this says that black holes are the most entropic entities we know of so a black hole about this big with a radius of about a meter as an entropy of about 10^{70} which is of course an astonishingly big and uh a black hole uh with the area of the Sun so a moderate sized cosmological black hole a real thing that we can observe with you know a gravity wave telescope or something has an entropy of about 10^{79} and really big black holes which are bigger than the entire solar system have entropies into the 10^{80} uh you know 10^{85} or something so these are enormously entropic entities but beckenstein didn't just tell us that he told us something about the structure of the interface The Horizon of a black hole and this is what he told us oh you can compute entropy and debts just

by using uh $\log_2$ instead of logs based e natural logs and that's just inter me multiplying the natural log by about 1.4 we can write the entropy of black hole of black hole in units of bits and it's about a over six so what does this mean to say that we can think of the entropy in terms of bits what it says is that we can think of the interface in terms of a bit array and we can think of all of these bits as encoded on this interface at a density of one bit every roughly uh six plank lengths of squared six plank areas so this is an incredibly dense encoding in a black hole oh and it's and it's an interesting idea about black holes but what gets really interesting is what happens when you generalize it and of course physicists are run to generalization and that's what happened next uh Gerard 't Hooft almost immediately thereafter on the basis that bekenstein's team's work formulated the holographic principle and what the holographic principle says is we can think of any system is approximately a black hole oh and the only approximation is the encoding density of the boundary of any system encodes the information that we can get about the system uh at some density and the density is less than the density for a black hole because we're not black holes uh and this rules out a lot by limiting the amount of information that you can extract from a system to the amount of information that you can actually write on its boundary and one of the things that it rules out is knowing the geometry on the inside of the system and to put it this way which I I think is a brilliant thought experiment he says look the metric inside this system can be so curved effectively the space time inside the system is can be so curved that you could stick an entire universe inside and we never know because it wouldn't change the amount of of entropy the number of bits that are encoded on the boundary so the inside geometry can be anything and you can't find out by looking at the outside of the system now the holographic principle almost follows from classical physics oh there's Euler's theorem which which tells you about $1/r^2$ forces and you can think of the forces penetrating the boundary at some density and the difference between Euler's theorem and the holographic principle is just that the density in classical physics can go to Infinity and in the holographic principle it can only go to this maximal density that's achieved by a black hole why because if you try to get any higher and more information the information gets sucked into the black hole and you can't get it out so the holographic principle becomes a guiding principle for thinking about any physical system and what it means to extract information from a physical system and in fact we'll talk about this next time you can say exactly what the information encoded on the boundary of any system is of and uh well I'll go through next time is is seeing that uh the information that's encoded on the boundary is a specification of the the energy that's being exchanged by the by the

interaction which of course is linear in the number of bits it's just counting the number of bits okay with the holographic principle we now have a complete new science about systems that are exchanging of finite discrete information across a boundary by encoding that information on the boundary and then reading the information off the boundary so here's a new way of thinking about physical interaction Alice writes a message on her boundary with Bob and Bob reads the message off the boundary and then writes his own new message which Alice then reads and of course that's what's happening right now I'm writing information by speaking on an interface which is effectively the internet and you're reading that information off the internet by listening to it and when we get time for questions you'll be writing information on the internet which I'll be reading now the key thing about this new science is that it's topological it's about connectivity across some Communication channel which we represent as a boundary it's not geometric so it doesn't assume anything about space-time so it allows us to build a model of space in particular as an emergent phenomenon and it allows us to seek time as not some absolute external abstraction but something that the communicating agents measure for themselves and so they each Act in their own time and we get a very natural measure of time in terms of how many bits I see coming across my boundary and again we'll talk about that in the next couple of sessions um and then talk about it more thoroughly in September when we talk about emergent SpaceTime so Quantum information Theory looks very different from the physics that came before not because it's adopted a new formalism of the tools of quantum theory it's because it's entirely changed the thinking about what physics is and what it's about oh and replace this idea of forces and balls banging into each other and all of that with the idea of communication between agents of course that's familiar from an active inference perspective so we can now uh back up a little bit to see what they were doing in classical physics during this period well let me let me go on a little bit sorry this is this is just a slight quoting uh wheeler who of course is the most radical in terms of empathiest in terms of formulating these ideas but here's his characterization of this new physics uh it's not reductive uh all of the reasoning is circular you don't have smaller things and then yet smaller things and then yet smaller things forever there are no laws so what you see on the interface uh is a is a message it's not something that's governed by laws since the beginning of the universe uh there's no Continuum so there's no there's nothing described by real numbers it's all described by integers everything's done in finite dimensional spaces and finally and most importantly there's no space-time box in which things happen so think of how radical this is means there's no big bang there's no big rip there's no bouncing universe all of those ideas are out the

window because they're classical and they're about space-time and what the new physics wants to do is derive space-time out of basically users experiences so the the agents here are all Observer participants in wheelers uh language but what that just means is Agents that that want to communicate and it's their communication that gives rise to so now we'll go back to classical and uh what was happening or one thing that was happening in classical physics at that time was a lot of thinking about uh stochastic causal Networks and um Pearl realize that if you have any stochastic causal Network that's unidirectional then around any node you can draw what he called a Markov blanket and a Marco blanket is just the set of nodes in the network that that absorb all outside causation and then transmit that causation into whatever node you're interested in and that absorb all causation coming from the node you're interested in and transmit it to the rest of the world and so we can redraw that in part D here and it should look very familiar a Markov blanket is just a classical physics way of talking about a holographic screen and the number of nodes in the Markov blanket or in particular the number of degrees of freedom times the number of nodes is just the number of bits that flow across that Markov blanket so it's the number of it's the entropy of the effective holographic screen so so this this all these ideas were reinvented more or less independently within classical physics and it was from this classical physics background that uh Carl Preston came up with the idea that a motor called blanket defines a persistent at least from an active inference Institute point of view everyone is familiar with because it's the foundation of the idea of active inference oh any system is that persists through time uh does so by making sure that it doesn't dissolve into its environment what does that mean it means it it persists Through Time by maintaining the Integrity of its Markov blanket or the Integrity of its boundary so this of course is just a tautology but it's a very it's a very interesting and very productive tautology because it says that any System is using the information it gets on its boundary from its environment to build a model of how its environment behaves and then it uses that model to act back on its environment to test and refine its model and again as popper to told us this is just what science is so what the free energy principle really tells us is that all systems are agents that are doing science all the time so physics is effectively uh not just the study of communication but it's the study of uh agents doing science with each other pairs of agents who are trying to figure each other out by their communication and communicative exchanges that's the history of how we got from 1930s sorry 1850s thermodynamics to the free energy principle how the free energy principle connects to these very deep and extremely radical especially within context ideas in quantum theory and Quantum cosmology and computer science all of which tell us the world

we see is a projection that's being written on our boundaries by a process that we have no access to accept the the procedure of active inference or the procedure of science which is to formulate predictive models and test them by doing things in the world and seeing how the world responds so uh that's it for this session the first discussion session which Anders going to lead will be the 3rd of June at this same time uh I.E five o'clock European Time and then uh my session number two will be in mid-june and uh if you look at the course website uh there's this subsidiary website for interactive q a and I invite everyone to post questions and uh discuss them and hope that we'll have an interesting Exchange and that [Music] everyone will come to sort of an understanding of what uh was talked about today through discussion and will be uh interested in being back in June to see how to formulate this in quantum theory so thank you very much and and thank you again Daniel for organizing this and hosting it and putting together all the technical things necessary to pull this off I could never do that on my own what thank you we're really um excited here any closing thoughts or or Andrew I'd love to hear your reflection just briefly what how would you have told that history or how does that history reflect on the areas that you're familiar with well I I don't think that had much to do on I I've been you know mainly interested in the week rotation part I I'm excited to hear more about that Chris uh but no I mean I I promise to be a very very nice summary I don't think I I'm less familiar with the with a stop on on computation formal computation despite being a mathematician by training um and and it was a little more familiar with with all the physics stuff but yeah I thought it was great thank you all right thank you I will uh close it so in about two weeks we'll have the first discussion everyone's welcome to join I will uh share with Hunter and Chris the questions and we can develop that and we'll have that on the course front end um before the coming discussion so thanks again fellows see you next time thank you